

Treatment of Severe Nodular/Conglobate Acne

High Strength of Recommendation

- **Oral isotretinoin** is strongly recommended as monotherapy for the treatment of severe nodular/conglobate acne due to its superior efficacy compared to systemic antibiotics or topical therapy alone.

Medium Strength of Recommendation

- **Systemic antibiotics** in combination with the fixed-dose combination of **adapalene and benzoyl peroxide (BPO)**, or with **azelaic acid**, can be recommended for the treatment of severe nodular/conglobate acne.

Low Strength of Recommendation

- **Systemic antibiotics** in combination with **adapalene** or **BPO** can be considered for the treatment of severe nodular/conglobate acne.
- For females: **Hormonal anti-androgens** in combination with systemic antibiotics and topical treatments (excluding topical antibiotics) can be considered.
- For females: **Hormonal anti-androgens** in combination with a topical treatment (excluding antibiotics) can also be considered.

Open Recommendation

- Due to insufficient evidence, no recommendation can be made for or against the use of **intense pulsed light (IPL)** or **laser therapy** in severe nodular/conglobate acne.
- Although **photodynamic therapy (PDT)** shows effectiveness, it cannot currently be recommended due to lack of standardized treatment protocols and concerns about acute adverse reactions.

Negative Recommendation

- **Topical monotherapy** is not recommended for conglobate acne.
- **Oral antibiotics** are not recommended as monotherapy.
- **Oral anti-androgens** are not recommended as monotherapy.
- **Artificial UV radiation sources** are not recommended.
- **Visible light therapy** as monotherapy is not recommended.

Choice of Topical vs. Systemic Treatment

There is limited evidence directly comparing topical and systemic therapies or evaluating the added benefit of combination therapy over topical treatment alone. Most studies compare topical antibiotic monotherapy with systemic antibiotic monotherapy.

In clinical practice, expert consensus supports that: - **Patients with severe papulopustular acne or moderate nodular acne** achieve better outcomes with **systemic antibiotic plus topical treatment**, or with **systemic isotretinoin**. - Involvement of the trunk is a key factor favoring systemic therapy. - **Systemic treatments** are associated with better adherence and higher patient satisfaction, particularly in widespread disease.

Additional Considerations

- **Isotretinoin** is highly effective in severe cases, supported by strong clinical experience despite limited controlled trial data specific to nodular/conglobate forms.
- Evidence for many treatment options was downgraded due to reliance on indirect data from studies of severe papulopustular acne, particularly when assessing nodule and cyst reduction.
- Prescribers should be mindful of the risk of **antibiotic resistance** when using systemic antibiotics.
- Systemic antibiotics such as **doxycycline** and **lymecycline** should be limited to a maximum treatment duration of **3 months**.

- In cases where higher-strength recommendations cannot be followed (due to cost, availability, licensing, or legal restrictions), lower-strength alternatives may be used as first-line therapy.

Treatment of Papulopustular Acne

There is insufficient evidence to recommend laser or light-based therapies (other than blue light) for papulopustular acne due to lack of standardized protocols, clinical trial data, and real-world experience.

Systemic treatment with **isotretinoin** is recommended in severe cases based on high efficacy observed in clinical practice, although safety and tolerability data remain limited and insufficient for guiding recommendations independently.